

Anderson Localization in a Ferrofluidic Waveguide System with a Topological Defect

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Abstract

Anderson Localization is a phenomenon negating wave diffusion in a spatially-disordered medium and currently is only a theoretical concept in Phononics. To attain true spatial distortion, the Phononic Crystal medium should not have a characteristic Bravais Lattice, which may be satisfied through a Fluid-Fluid Ferrofluid medium in a magnetic field. Further enhancement of Anderson Localization effects may result from a topological defect allowing for Soliton propagation. Thus, the purpose of this project is to develop a Ferrofluidic waveguide system with a topological defect under a controlled Magnetic Field to physically observe Anderson Localization. The success of this project will be determined via simulated degrees of Acoustic Pressure damping (using COMSOL® Multiphysics 5.6). The experimental setup was created through a CAD kernel, simulating Magnetic Flux Density and Total Acoustic Pressure. Nine periods of N52-model magnets were placed along edges of a 400-millimeter air-environment block to form a periodic structure (control); a vertical deviation of $\frac{25}{6}$ millimeters in the 5th and 6th magnet periods formed the topological defect. Compared to the sinusoidal Total Acoustic Pressure plot of the control, the defect-induced Total Acoustic Pressure plot experienced trigger peak damping, demonstrating the ultrasound-isolation effect of Anderson Localization. The use of commercially-available components further offers practicality. Therefore, the Ferrofluidic waveguide system developed successfully demonstrates the effects of Anderson Localization in Phononics and can be physically realized. The setup realized may be applied to future Phononic localization effects and can aid in the practical optimization of speaker systems using Ferrofluids.

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Introduction

Subject

In the subfield of Phononics in Acoustic Physics, vibrations in lattice structures are studied to observe various phenomena. These lattice structures, called crystals, are classified into three categories: Solid-Solid, Solid-Fluid, and Fluid-Fluid Phononic Crystals [1]. A unit cell of a Phononic Crystal typically has a characteristic shape, according to its classification under Bravais lattices [2]. Removing one cell from the crystal or altering the period of repetition between cells is equivalent to distorting the lattice structure, which can cause a topological defect.

A topological defect is a special case of a Soliton, which can be thought of as a stationary localized wave that propagates through space yet does not move forward [3, 4]. In theory, a soliton, once disrupted, should be able to recover its original state of stasis.

In 1958, American Physicist P.W. Anderson reported a strong localization effect that occurred in disordered crystals and resulted in the absence of wave diffusion [5]. This phenomenon of Anderson Localization hence has the potential to realize sound-isolation upon the propagation of a sound wave in a sufficiently disordered crystal. Existing literature in Photonics, the light-counterpart of Phononics, has described experimental setups [6] where superlattice structures [7], optical fibers [8], and semiconductor powders [9] have been used to demonstrate Anderson Localization. In Phononics [10], the use of physical scatterers has been theorized with concepts such as suspended air bubbles through a fluidic medium [11], random-sized spherical scatterers [12], and randomly located 2-dimensional beams [13]; while these concepts work in theory, it is hard to physically construct these setups for experimentation. In fact, such physical scatterers are typically constructed with metal and hence can introduce error in the experimental setup with unintended effects of corrosion.

One experimental method has been reported to demonstrate the effects of Anderson Localization without the use of physical scatterers [14]. Without the use of artificial scatterers to distort the medium, the medium must be spatially distorted while maintaining a stable state. One way of achieving this is to manipulate a Fluid-Fluid Phononic Crystal as if its nanoparticles were solids, which is a method suitable for Ferrofluids [15]. The concept of Ferrofluids was an attempt by the National Aeronautics and Space Administration (NASA) to guide the flow of rocket fuel in a zero-gravity environment. While Ferrofluids were ultimately not used for their intended purpose in jumpstarting rocket engines, they have since become commercially popular due to their solid- and liquid-like characteristics. Introducing a magnetic field to Ferrofluids will cause the nanoparticles

to act as individual point objects as opposed to the fluid reacting as a whole [16, 17].

Purpose

The purpose of this research study is to create a Ferrofluidic waveguide system using solenoids and/or magnets with a topological defect to enhance the effects of Anderson Localization on the Ferrofluid. Anderson Localization, in idealized conditions, should negate all wave propagation and hence would allow a system to become a sound-isolation chamber. While this effect will not be achieved in reality, its potential to be realized can be maximized via the introduction of a topological defect and sufficient spatial distortion from the controlled magnetic fields on the Ferrofluid. Doing so will provide an experimental method to validate Anderson Localization in a realistic setup.

Scope

The success of this research investigation will allow for Anderson Localization to be more concretely observed, which will enrich one's understanding of the physical processes that occur in a medium while Anderson Localization is achieved. Furthermore, this study will provide a method for Anderson Localization to be practically applied in the real world, enhancing the commercial market for speakers, sound isolation systems, etc. As these devices typically use Ferrofluids, applying – or avoiding – the effects of Anderson Localization can be put into place with relative ease, improving both scientific and commercial understanding of practical localization effects.

Development

The methods followed throughout this investigation will be presented first to provide the reader with a foundation on creating simulations through the COMSOL® Multiphysics 5.6 software. Any assumptions made that may impact the study will be discussed initially. The general methodology followed will be listed before the results of this study to guide the reader through the research investigation. The results will be accompanied with an analysis of the simulations conducted under the 'Discussion' section to justify the success of the waveguide model and will additionally allow reconstruction of this research study by the reader. Lastly, conclusions will be drawn from the waveguide apparatus created, and references will be shown to enhance the reader's comprehension on the subject matter.

Methodology

Methods

All simulations in this study were conducted using the COMSOL® Multiphysics 5.6 software. For consistency, the metric system was followed in all measurement quantities.

In all simulations, an environment of air was created to be of dimensions 100 millimeters by 100 millimeters by 400 millimeters. Similarly, the acrylic plastic waveguide was consistently a hollow cylindrical tube with an outer radius of 26 millimeters, an inner radius of 25 millimeters, and a height of 400 millimeters. In initial designs, the Magnetic fluid (modeled by a cylinder of radius 25 millimeters and height 400 millimeters) inside the waveguide was set to be liquid Neodymium for convenience; later designs used Ferrotec’s commercial-grade EFH1-model Ferrofluid.

In design models with solenoids, the Copper coils (helices) had a major radius of 26.2 millimeters and a minor radius of 0.1 millimeters; in addition, each period of coils contained 150 turns and separate periods of coils were connected with a single-turn coil. Upon completion of constructing the 9 periods of coils, a Composite Domain virtual operation was formed to join all periods of coils into a single solenoid. Then, an extrude operation would be performed on the ends of the completed solenoid to touch the exterior of the air environment block, and an electrical current would be run through the solenoid via the coil operation under the Magnetic Fields interface.

Design models with Neodymium permanent magnets used cubes of edge length 25 millimeters, situated at the boundaries of the air environment block, to simulate a controlled magnetic field. Similar to the solenoid models, nine periods of magnets were used; hence, four magnets would each be positioned at the same height, and each period of magnets would have a vertical separation of $\frac{50}{3}$ millimeters in between faces. These Neodymium magnets were placed under a Magnetic Flux Conservation constraint with a Magnetization of $1000 \frac{\text{A}}{\text{m}}$ in the x -, y -, and z -directions.

When producing the topological defect, a Parametric Sweep on a variable parameter p was used to adjust locations of the magnets in the middle (namely, the fifth and sixth set of magnets, out of the ten sets of magnets). The following scheme was used to position the magnets:

All Simulations created used the Pressure Acoustics, Frequency Domain 3-dimensional model. Simulations with the solenoids additionally used the Magnetic Fields model, whereas simulations with the permanent magnets used the Magnetic Fields, No Currents model. Magnetic Flux Density Volume and Slice 3-dimensional plots were created with a selection set of the Magnetic-Field-inducing agent (i.e. the solenoids or permanent magnets) and the inner fluid of the waveguide. The

Magnet Set	Magnet Center of Mass Height	Gap Center of Mass Height
1	12.5	$33.\bar{3}$
2	$54.1\bar{6}$	75
3	$95.8\bar{3}$	$116.\bar{6}$
4	137.5	$158.\bar{3} + \frac{p}{2}$
5	$179.1\bar{6} + p$	200
6	$220.8\bar{3} - p$	$241.\bar{6} - \frac{p}{2}$
7	262.5	$283.\bar{3}$
8	$1304.1\bar{6}$	325
9	$345.8\bar{3}$	$366.\bar{6}$
10	387.5	

Table 1: Magnet and Gap Center of Mass Locations

Acoustic Pressure Volume Plots were created with a Plane Wave propagating in the z -direction in the waveguide. To observe the effects of various degrees of the topological defect, 1-dimensional plots of Magnetic Flux Density versus Arc Length (height) and Acoustic Pressure versus Arc Length were created, both with respect to a cut line across the center of the waveguide.

Procedures

Initially, existing literature was reviewed to gain insight in current experimental methods of observing Anderson Localization, background in relevant Acoustic concepts, and ideas on creating/improving designs for an experimental setup for Anderson Localization. Upon completion, the COMSOL® Multiphysics 5.6 Software was explored to identify requirements and methods needed to create a waveguide system. To that end, the methodology behind creating plots and displaying relevant data was learned and eventually applied to carry out this research study. Finally, the following technical report was written to convey all results obtained during the duration of this study in a coherent manner.

Results

Solenoid Approach



Figure 1: Existing Physical Waveguide Model

Initially, the goal was to create a model of a waveguide similar to the one above on the COMSOL® Multiphysics 5.6 Software. The model needed to have nine periods of Solenoids (i.e. ten Solenoids), with each Solenoid composed of a 150-turn Copper Coil and a single-turn Copper Coil connecting periods. To effectively design a waveguide model, smaller simulations were conducted to analyze the Magnetic Flux Density of Solenoids.

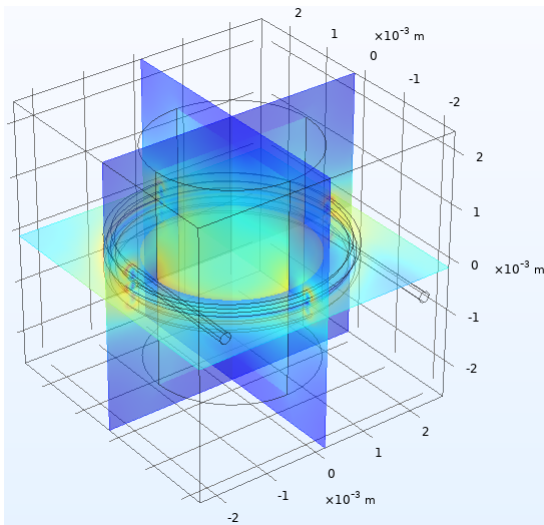


Figure 2: Magnetic Flux Density of a 2.5-turn Coil acting as a Solenoid

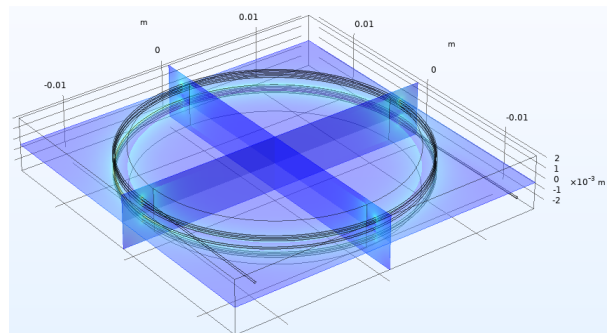


Figure 3: Magnetic Flux Density of Two 2.5-Turn Coils Connected by a 1-Turn Coil

It was observed that the far side of the coil (the locus of points on the coil surface that appeared closest to the waveguide) had the strongest Magnetic Flux Density. The same Coil Geometry

Analysis study was run on a miniature waveguide system (Figure 3), which demonstrated similar properties of the coil. As a result, the COMSOL® Multiphysics 5.6 Software treats the solenoids as uniform magnetic-field-inducing agents in an idealized setting, which behave no differently than the individual coils. This observation allows us to eventually scale our model into an experimental setup as it shows versatility in its application in the real world.

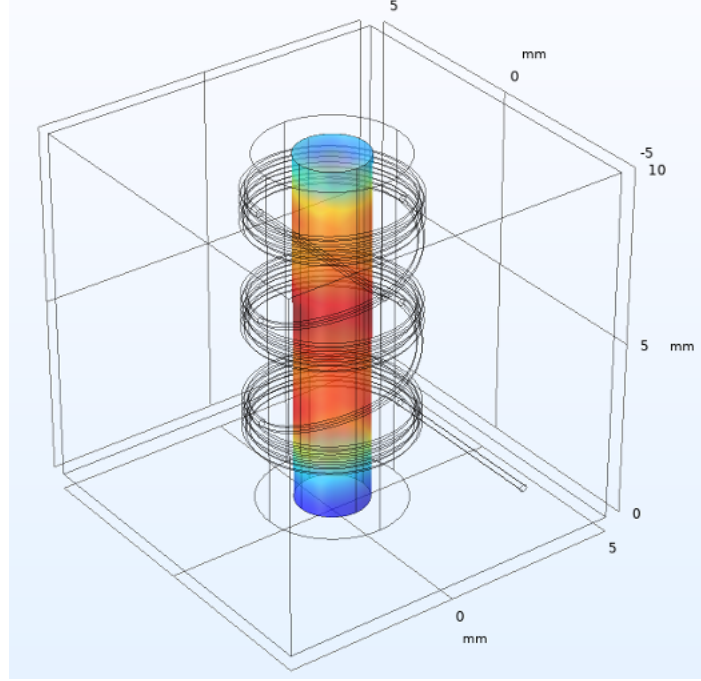


Figure 4: Magnetic Flux Density Volume Plot on a Ferrofluid Selection Set From a Solenoid

The Ferrofluid inside the waveguide can be considered as a Fluid-Fluid Phononic Crystal. While inducing spatial distortion in the Ferrofluid, locations of higher distortion degrees must be identified. In Figure 4, an even period of solenoids is present (i.e. an odd number – 3 in this case – of solenoids), indicating that the Magnetic Flux should coalesce at the center (to form a relative maximum), which is observed. With an even period – odd number – of solenoids, the Magnetic Flux should again have a relative maximum at the center. In both cases, the center of the waveguide is the location of local extrema, demonstrating that it is an apt location for eventually placing a topological defect. Inducing a topological defect will increase the efficiency of the sound-isolation chamber (i.e. the waveguide system) as it will theoretically negate all sound by allowing a Soliton to propagate in space without moving forward.

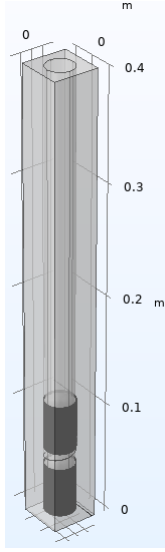


Figure 5: Waveguide Model Simulation

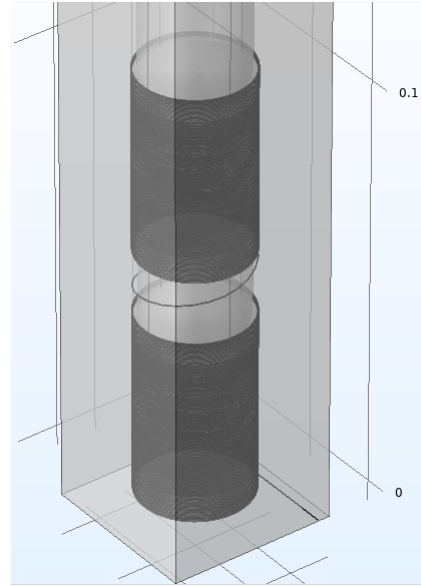


Figure 6: Section View of the Waveguide Model

In Figure 5, a full-scale waveguide system model can be observed with only one period of solenoids. Due to the small size of the coils relative to the waveguide, a Free Tetrahedral mesh of the Extra Fine size was necessary to construct a proper mesh without self-intersections along the system. As a result, running the Coil Geometry Analysis on just two solenoids exhausted the memory of the computer and pushed the COMSOL® Multiphysics 5.6 Software to use the MUMPS out-of-core memory space; however, the computer used did not have the required computational power to execute the analysis. Time and computational-power constraints further rendered this waveguide model obsolete.

Permanent Magnet Approach

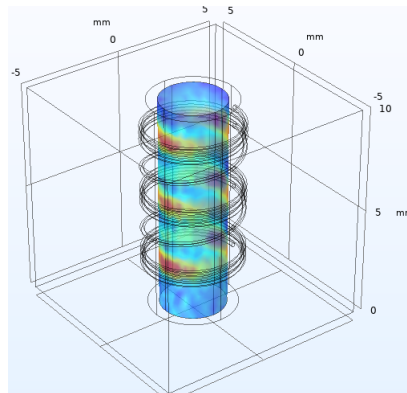


Figure 7: Magnetic Flux Density Volume Plot on a Ferrofluid Selection Set From Magnetic Coils

Figure 7 highly resembles Figure 4. However, while the earlier plot was made with the Magnetic Fields interface on the COMSOL® Multiphysics 5.6 Software and is a proper solenoid, the above Magnetic Flux Density plot is made with the Magnetic Fields, No Current interface. Hence, the coils are not solenoids, but act as permanent magnets. Intuitively, this substitution makes sense as solenoids produce a uniform magnetic field upon receiving a current, and permanent magnets will fundamentally produce a uniform magnetic field. Visual observation of figures 4 and 7 further justify the use of permanent magnets as they have little to no differences from the Solenoids and eliminate the Coil Geometry Analysis study step (which will save real time and memory/computational power). Hence, the waveguide model can be switched to using permanent magnets in hopes of progressing the research study.

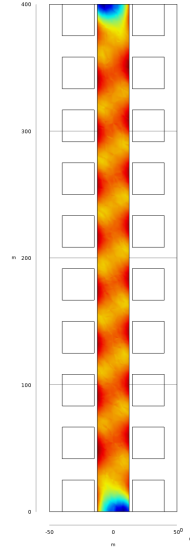


Figure 8: Magnetic Flux Density Slice on a Ferrofluid Selection Set From Permanent Magnets

The use of Neodymium permanent magnets in this manner creates visible bands of Magnetic Flux since there is a gap present between periods of magnets; with the use of solenoids, all coils were connected, which eliminated the repeated bands and instead created one coherent band pattern. However, this does not pose a problem since the center of the waveguide remains as the location of extrema, allowing for the procedure to continue with the idea of a topological defect.

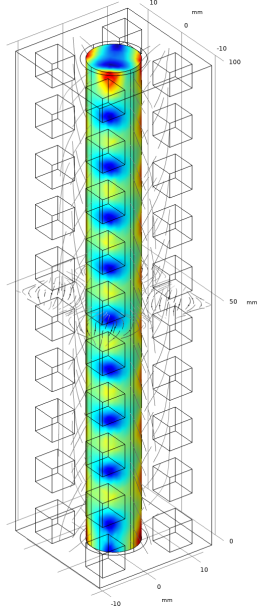


Figure 9: Magnetic Flux Density Volume Plot on a Waveguide Selection Set

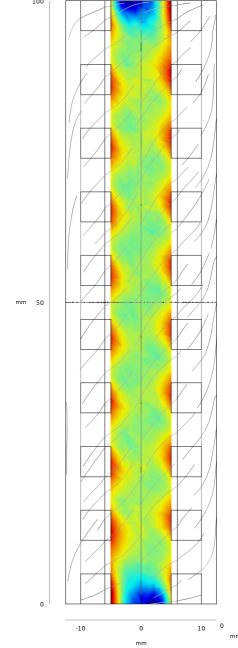


Figure 10: Magnetic Flux Density Slice on a Waveguide Selection Set

The presence of the so-called waveguide shell (acrylic plastic hollow cylinder) dampens the Magnetic Flux Density of the Ferrofluid, which is to be expected as acrylic plastic is a non-magnetic material. While this makes the interior Magnetic Flux to be uniform, it introduces a stronger contrast between the Magnetic Flux Density of the interior of the Ferrofluid and the Ferrofluid along the surface of the tube, which will only aid the concept of the topological defect.

Now that the Magnetic Flux Density studies are completed, the Acoustic Pressure Plots can be created. As the Acoustic Pressure plots require spatial distortion of the Ferrofluid, the Magnetic Flux Density plots needed to be created prior to running the Acoustic Pressure studies on the COMSOL® Multiphysics 5.6 Software.

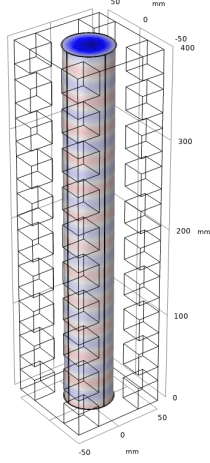


Figure 11: Acoustic Pressure Volume Plot for a Periodic Structure of Magnets

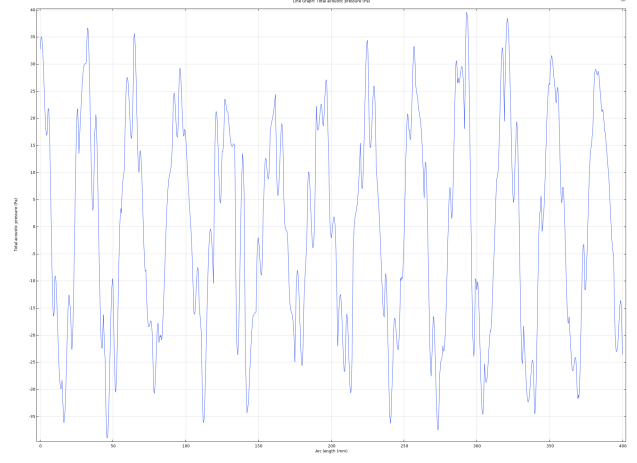


Figure 12: Line Graph of Acoustic Pressure Through the Ferrofluid for a Periodic Structure

Figure 12 shows the Acoustic Pressure through a vertical cut line passing through the center of the cylindrical Ferrofluid-and-waveguide system. One can observe that the Periodic Structure of Magnets has a lot of oscillations on its Acoustic Pressure, but acts as a uniform sinusoid and hence shows that the Periodic Structure achieves a very small degree of sound isolation.

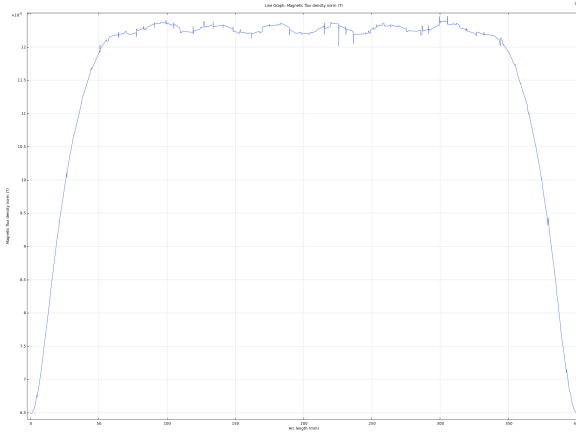


Figure 13: Line Graph of Magnetic Flux Density Through the Ferrofluid for a Periodic Structure

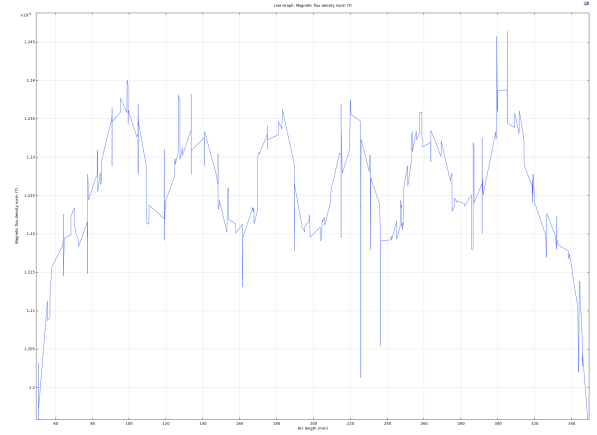


Figure 14: Section View of the Magnetic Flux Density Line Graph for a Periodic Structure

Figures 13 and 14 show that the Magnetic Flux Density can be fitted to a sinusoidal curve for a Periodic structure of magnets. Hence, it is expected that the topological defect will break this sinusoidal pattern and will elongate the curves to a certain degree depending on the value of the parameter p (recall Table 1). Figure 13, being a Periodic structure, had $p = 0$. As the vertical separation between the faces of two sets of magnets is $\frac{50}{3}$, the value of p is restricted to $0 \leq p \leq \frac{25}{3}$. To gain a solid understanding of the changes in the Magnetic Flux Density plots, three studies can be run simultaneously: $p = 0$, $p = \frac{25}{6}$, and $p = \frac{25}{3}$.

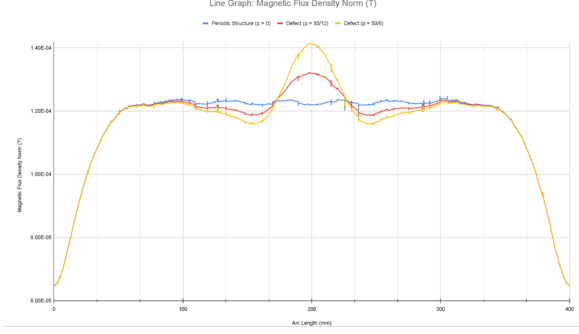


Figure 15: Line Graph of Magnetic Flux Densities for $p = 0$, $p = \frac{25}{6}$, and $p = \frac{25}{3}$

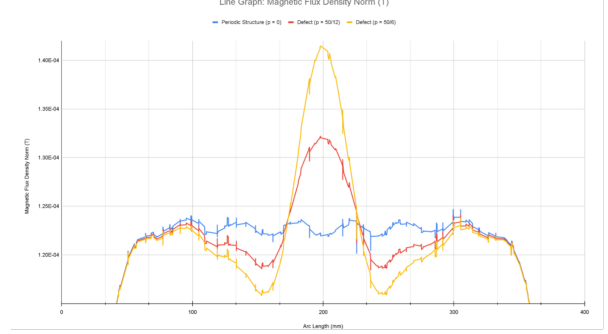


Figure 16: Section View of Magnetic Flux Densities for $p = 0$, $p = \frac{25}{6}$, and $p = \frac{25}{3}$

As p approaches $\frac{25}{3}$, the Magnetic Flux Density plot will slowly converge from the blue plot to the red plot and, eventually, to the yellow plot. The red plot appears to be the most stable and could be the best location for the topological defect; further tests of the Acoustic Pressure Plots will confirm the occurrence of a topological defect if the Acoustic Pressure decreases rapidly and becomes comparatively negligible to the readings of the trigger pulse.

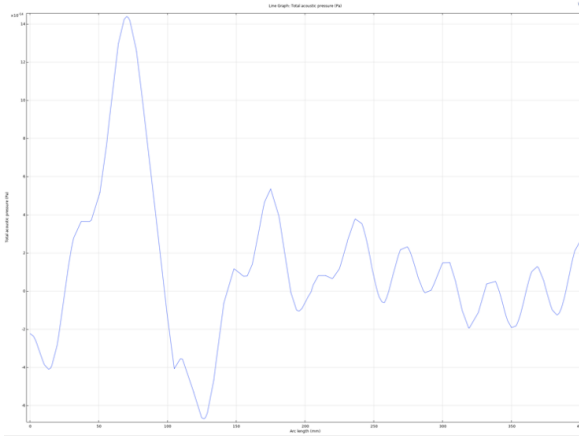


Figure 17: Line Graph of Acoustic Pressure Through the Ferrofluid for $p = \frac{25}{6}$

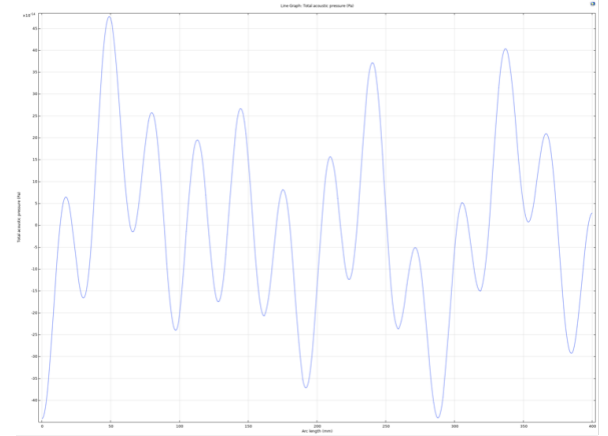


Figure 18: Line Graph of Acoustic Pressure Through the Ferrofluid for $p = \frac{25}{3}$

The Acoustic Pressure plot for $p = \frac{25}{6}$ millimeters shows the decrease of the Acoustic Pressure as the wave propagates to the end of the waveguide. Hence, maintaining a deflection of $p = \frac{25}{6}$ millimeters demonstrates the intended effects of Anderson Localization and provides theoretical results close to ultrasound isolation.

Discussion and Conclusions

Discussion

The findings of this research study are based in an idealized environment, created via the COMSOL® Multiphysics 6.1 Software. While this enables results to appear almost hyper-accurate, it introduces bias when extrapolating the results discussed to realistic setups. Thus, the results indicated in this report, while promising in its ability to be closely replicated in experiment, are prescriptions for experimental results.

However, the design modeled has great versatility for application in experimental setups. An acrylic plastic cylindrical waveguide coupled with Neodymium magnets or Copper coils (for the solenoid models) are relatively accessible compared to specialized equipment created for a specific method of experimentation. As a result, the design created has great practicality and can be used as a standard waveguide for experiments regarding localization, or more broadly, requiring a distorted Phononic Crystal.

During the testing of simulations on the COMSOL® Multiphysics 5.6 software, the computational power and memory space required for thorough functionality exceeded the limits of the device used. Hence, the MUMPS-out-of-core module was used for extra memory. It is unknown whether this action introduced bias in data filtration, but more-conclusive simulations would require the use of a supercomputing device for its computational power and memory.

During the creation of the waveguide model with solenoids, self-intersections would occur in the mesh. To keep the Free Tetrahedral mesh, a smaller size of mesh (Extra Fine) was required; however, this mesh scaled the number of data points almost ten-fold, which was not sustainable with a deficit of computational power. To work around this, the Free Tetrahedral mesh was swapped for a Free Triangular mesh; as the Free Tetrahedral mesh is more accurate for data optimization purposes, the Free Triangular mesh may be considered as a limitation to the degree of accuracy presented in this research study.

While Ferrotec’s commercial-grade EFH1-model Ferrofluid was selected to be used, only generic material properties (e.g. density, viscosity, appearance characteristics) were reported. As Acoustic-related material properties were needed, some were obtained through a literature review and others were fixed to be relatively convenient values for the COMSOL® Multiphysics 5.6 software. While this may be regarded as a limitation, it has negligible impacts on the study when considering normalized (magnitude) plots.

Similar to an experiment, repeated testing of the same simulation would yield similar values in the Acoustic Pressure Plots; occasionally, the plots would return with significantly varied results, which were deemed as erroneous plots resulting from error present in the study. While actual experimentation was not conducted, the simulated results of this research study are promising and hence are expected to scale to a real-world experimental setup.

Conclusion

The Ferrofluidic waveguide system included a successful topological defect with nine periods of Neodymium magnets along the corners of the air environment block. When the fifth and sixth set of magnets were moved $p = \frac{25}{6}$ millimeters up and down, respectively, to introduce the topological defect, the Acoustic Pressure visibly approached zero Pascals at the end of the waveguide compared to the trigger pulse at the start of the waveguide. Hence, the waveguide system developed has the potential to be used in an experimental setup to observe the effects of Anderson Localization in an ultrasound isolation chamber (i.e. the waveguide system). Furthermore, the model can be created physically with relative ease as it does not have customized parts, allowing for this model to have versatility in physical applications. Thus, the waveguide system developed not only is efficient in achieving Anderson Localization in simulations, but also has the potential to be efficient in physical observation of Anderson Localization and related phenomena.

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